

Retrieval Augmented Generation

Introduction

Retrieval Augmented Generation, commonly called RAG, combines information retrieval with large language models. Instead of relying only on the knowledge stored in the model, RAG retrieves relevant information from external documents.

Vector Search

Vector search converts text into numerical embeddings and finds documents that are semantically similar to a user question. It is useful when the exact words in the question are different from the words in the document.

Keyword Search

Keyword search finds documents using exact or closely matching terms. BM25 is a popular keyword retrieval algorithm. Keyword search is especially useful for names, identifiers, product codes, and technical terms.

Hybrid RAG

Hybrid RAG combines vector search and keyword search. Vector search provides semantic understanding while keyword search provides exact term matching. The results can be combined before sending the retrieved context to the language model.